ABACUS

Aircraft engine analysis and experimental ML

Final software handover report | 18 September 2026

Academic steady-state simulation, database and surrogate modeling. Independent real-engine validation remains incomplete.

1. Scope and delivered outcome

The project compares 13 named engines across turbojet, high- and low-bypass turbofan, turboprop and piston architectures. Inputs are altitude, Mach number and throttle. The physics backend predicts thrust or shaft power, fuel flow and architecture-appropriate fuel-consumption metrics. Experimental ML approximates this backend within saved input bounds.

The delivered archive includes the supplied working database, source code, trained models, historical execution reports, calibration candidates and a portable launcher. It preserves the original engine parameters and operational ML validation gates.

Area	Evidence and status
Database	13 engines; integrity OK; no foreign-key violations; 218 numerical parameters match provenance.
ML and common comparisons	13 selected model files present; recorded unified matrix: 780/780 available; common conditions: 78/78 available.
Turboprop calibration	Two separate single-point fitted candidates, with 60-condition comparisons per engine. Independent validation pending.
Simulink	User-reported targeted parity test: 1 passed, 0 failed, 0 incomplete. No transient dynamics claim.
Handover	New launcher and GUI integration changes require the supplied MATLAB smoke check.

Evidence boundary: archive/CSV/database checks were performed during packaging. Historical MATLAB outcomes come from saved reports or the user's console transcript. MATLAB was not available in the packaging environment; no new MATLAB execution is claimed.

2. Method and experimental design

The atmospheric model supplies operating conditions to a shared gas-turbine cycle for turbojets, turbofans and turboprops, with a separate piston model. The turboprop cycle balances compressor work, then extracts remaining gas energy through a power turbine and gearbox. These are simplified steady-state models with estimated/assumed inputs; they are not manufacturer engine maps.

The original ML run generated 720 points per engine, with 503 training, 108 validation and 109 test rows. Inputs varied jointly; the fixed-Mach, fixed-throttle altitude plots are slices through the model and do not describe the entire training dataset. Subsequent coverage and boundary studies challenged the initial models.

Study	Requested	Accepted / available	Interpretation
Unified prediction matrix	780	780	Availability only, not accuracy.
Six common conditions x 13	78	78	Comparable query conditions.
Mach sweep	273	170	103 rejections explicitly cite configured Mach limits.
Throttle sweep	143	131	12 cycle failures: 8 nozzle pressure, 3 power turbine, 1 net thrust.
Calibration comparison	240 result rows	240	120 conditions, each evaluated with original and candidate.

The calibration comparison uses altitude {0, 2000, 4000, 6000, 8000} m, Mach {0, 0.2, 0.4, 0.6}, and throttle {0.5, 0.75, 1}: 60 combinations per engine. The sea-level static full-throttle point is the calibration point and must not count as independent validation.

Rejection reasons describe the implemented model and configured envelopes. They do not establish actual engine operating limits. Future MATLAB sweep plots now retain rejected rows as NaN gaps; historical PNGs are preserved as historical outputs.

3. Surrogate model evidence

Engine group	Selected model
J79-GE-17A	Reviewed Chebyshev model; TSFC derived from predicted fuel flow / thrust.
Atar 9C, R-25-300, F100-PW-229	GPRLog boundary revisions.
CFM56, CF6, M88, Kaveri, three turboprops, two pistons	Original experimental polynomial bundles.

The earlier J79 GPR boundary revision remained REVIEW_REQUIRED and is not selected. The later Chebyshev study reported 640 interior, 300 face and 120 edge points, plus eight seen training corners. Corner checks are not fresh hold-out evidence.

J79 cohort	Target	Relative RMSE (%)	MAPE (%)
Interior	Thrust_N	0.140	0.113
Interior	FuelFlow_kgs	0.112	0.095
Interior	TSFC_kgNs	0.062	0.030
Face	Thrust_N	0.205	0.148
Face	FuelFlow_kgs	0.155	0.121
Face	TSFC_kgNs	0.110	0.045
Edge	Thrust_N	0.106	0.097
Edge	FuelFlow_kgs	0.067	0.076
Edge	TSFC_kgNs	0.446	0.041

These metrics compare the surrogate with synthetic simulator outputs. They do not measure errors against real engines. The piston constant-target R2 review remains a reporting caveat: a nearly constant target should be assessed using meaningful absolute error and variation, rather than interpreting R2 alone.

Original training logs reported raw prediction speedups of roughly 5.55-21.72x. Those timings do not benchmark the packaged GUI: its guarded predictor also runs a physics validity check. No new latency benchmark was performed.

4. Turboprop calibration findings

Engine	Original airflow	Effective fitted airflow	Original / fitted power
PW127G	16 kg/s	8.470194 kg/s	4114.19 / 2178 kW
AE 2100D3	26 kg/s	12.244267 kg/s	7342.87 / 3458 kW

The fitted airflow equals original airflow multiplied by target power / baseline power. Each fit matches its chosen sea-level static full-throttle target to numerical precision. Effective airflow is an adjustable model parameter here, not an independently measured mass-flow value.

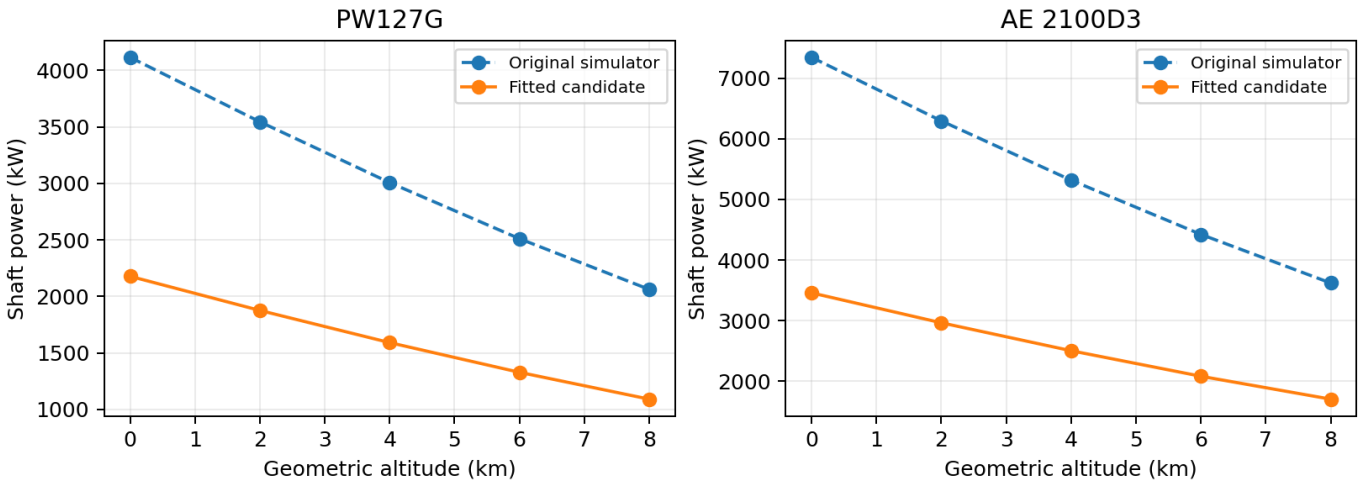


Figure 1. Simulator predictions at Mach 0 and throttle 1. Only the sea-level candidate point was fitted to the chosen rating. All other plotted points are model extrapolations from that calibration, not independent measurements.

Across all 60 conditions per engine, power and fuel flow scale by 0.529387 for PW127G and 0.470933 for AE 2100D3. BSFC and efficiencies remain unchanged to floating-point precision. Thus the fit corrects the output scale at one point, without independently confirming fuel consumption or off-design behavior.

The database records PW127G 2178 kW as a reviewed EASA published rating, with dry sea-level static conditions. AE 3458 kW remains provisional. The fitted candidates retain Low confidence and are stored separately; original database parameters and registered ML models are unchanged.

5. Validation, limitations and traceability

Track A compares physics predictions with reference data. The saved validation report contains 13 WARNING rows and one verified-reference FAIL: the original PW127G model predicts 4114.19 kW against 2178 kW, an 88.898% error. Its fitted candidate is not a new independent validation pass. AE's original discrepancy against the provisional 3458 kW target is 112.34%.

Other reference warnings include unverified sources and an afterburner case outside implemented scope. Source verification and model agreement are separate questions. Internal conservation residuals and unit tests cannot establish that assumed component inputs represent a real engine.

Remaining limitation	Required evidence or action
Real-engine accuracy	Independent, condition-matched manufacturer/test measurements across altitude, Mach and throttle; fuel flow as well as power/thrust.
Calibration identifiability	Justify effective airflow and uncertainty; account for interactions with turbine temperature, efficiencies and control schedules.
AE target	Resolve provisional rating and unit/operating-condition interpretation before treating the fit as authoritative.
Operating envelope	Validate actual limits and part-load behavior; do not treat simulator rejection thresholds as certification limits.
Candidate ML	Original ML does not represent fitted candidates. If candidate surrogates are needed, train/version them separately and preserve independent tests.
Simulink scope	Current models wrap steady-state backend calculations. Transient spool dynamics are outside the demonstrated scope.

Evidence files are retained beneath Aircraft_Engine_Digital_Twin/reports. The evidence index lists their exact paths. The package SHA256SUMS.txt records all distributed files except itself. The original archive and supplied working database were the source of this handover; changes are listed in PACKAGE_CHANGES.md.

Conclusion: this is a functioning academic software prototype with documented simulator-surrogate tests and an experimental calibration study. It must not be described as a manufacturer-validated digital twin. Independent engine validation remains future engineering work.

6. Installation and handover procedure

1. Extract the complete ZIP into a new writable folder. Keep the existing working project as a backup. Restart MATLAB to avoid old class definitions and paths.

2. In MATLAB, open the extracted Rebuit folder containing LAUNCH_ABACUS.m. Do not add the entire Rebuit tree recursively to the path; historical add-ons contain duplicate function names.

3. Run this short integration check in the Command Window:

```
[~, registry, smoke] = LAUNCH_ABACUS('check');
```

The launcher checks all 13 fingerprints, writes relocated bundle copies into a new reports subfolder, and queries each engine at [3000 m, Mach 0.3, throttle 0.8]. Expected success is 13 available predictions. This is a portability check, not additional training or manufacturer validation. If a fingerprint differs, stop and inspect the reported engine; do not bypass the guard.

4. Open the application:

```
[app, registry] = LAUNCH_ABACUS('gui');
```

Use the engine dropdown inside Experimental ML. The top main-GUI engine selector is independent. The launcher supplies the registry and predictor automatically. Input order for unified command-line predictions is [geometric altitude in m, Mach, throttle].

5. Example query:

```
prediction = PREDICT_UNIFIED_ENGINE(...
registry, 1, [3000, 0.4, 0.8]);
```

The historical J79 result at this point is approximately 30094 N thrust and 0.88612 kg/s fuel flow. Exact agreement must be checked in MATLAB after relocation. Queries outside saved bounds remain unavailable; there is no silent experimental-model substitution.

Environment used by the user: MATLAB R2026a Update 4 on Windows 11, with Database Toolbox, Statistics and Machine Learning Toolbox, and Simulink available. Simulink is needed for Simulink tests; saved GPR models require the statistics toolbox. The launcher requires JVM support for fingerprints.

No full retraining or long sweep is needed for the launch check. Existing study scripts retain their historical file-selection workflows; the launcher covers the packaged GUI and unified predictor. For reproducibility of older studies, use the evidence index and retained source scripts.